

Please stick the barcode label here.

Candidate Number

BIOLOGY PAPER 1

SECTION B: Question-Answer Book B

This paper must be answered in English

INSTRUCTIONS FOR SECTION B

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5 and 7.
- (2) Refer to the general instructions on the cover of the Question Paper for Section A.
- (3) Answer **ALL** questions.
- (4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (5) Supplementary answer sheets will be supplied on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.
- (6) Present your answers in paragraphs wherever appropriate.
- (7) The diagrams in this section are **NOT** necessarily drawn to scale.
- (8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.



SECTION B

Answer **ALL** questions. Put your answers in the spaces provided.

1. For each of the brain parts listed in column 1, select from column 2 one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1

Cerebellum

Medulla oblongata

Cerebrum

Column 2

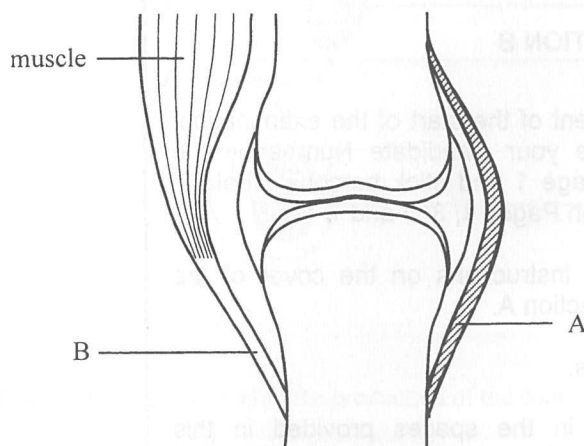
A. Damage causes inability to make decisions

B. Damage causes loss of body balance

C. Damage causes loss of control of breathing rate

D. Damage causes loss of withdrawal reflex

2. The diagram below shows the structures of a joint.



- (a) Name the type of joint shown and suggest an example of such a joint in the body. (2 marks)

Type of joint:

Example in the body:

- (b) Briefly describe how A and B work together to bring about movement at the joint shown. (4 marks)

Answers written in the margins will not be marked.

3. Figure A shows a section of part of the human alimentary canal. Figure B shows another section of the same part with blood vessels stained.

Figure A

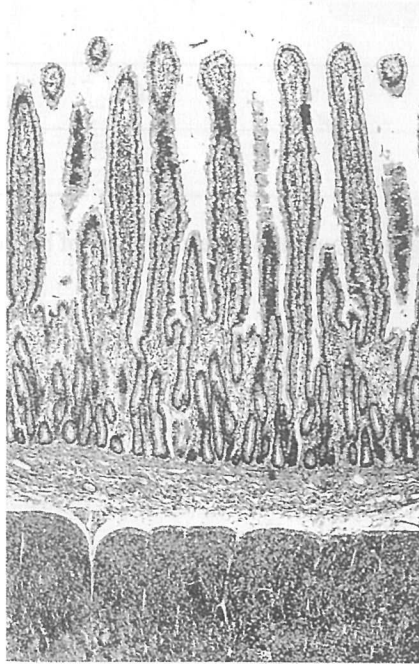
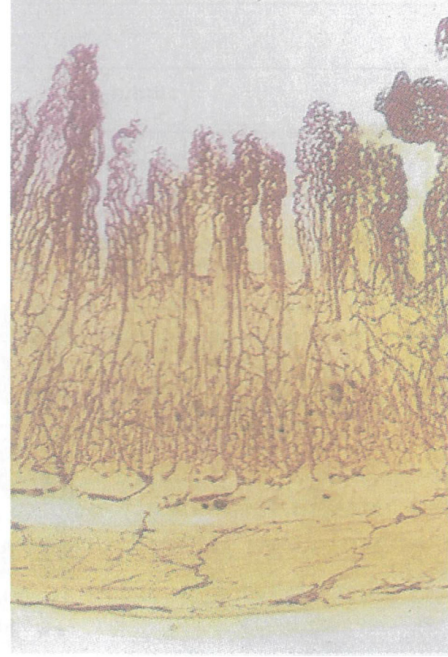


Figure B



- (a) Which part of the alimentary canal is shown in the figures? Support your answer with evidence. (2 marks)

- (b) For each of the above figures, describe *one* observable feature and explain how it is related to the functioning of this part of the alimentary canal.

Figure A: (2 marks)

Figure B: (3 marks)

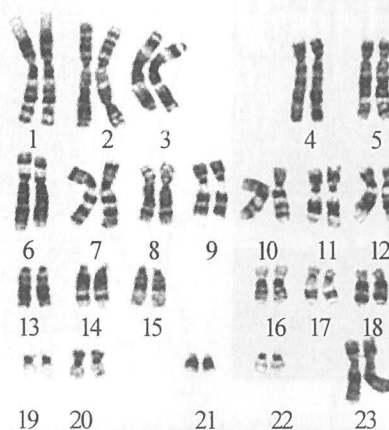
Answers written in the margins will not be marked.

4. Red-green colour blindness is an X-linked recessive trait in humans. Peter is red-green colour blind while his daughter, Mary, is normal.

(a) Deduce Mary's genotype without using a genetic diagram.

(4 marks)

(b) Mary is an expectant mother. The photomicrograph below shows the karyotype of her foetus:



- (i) From the photomicrograph, can we deduce whether this foetus will be red-green colour blind or not? Explain your answer.

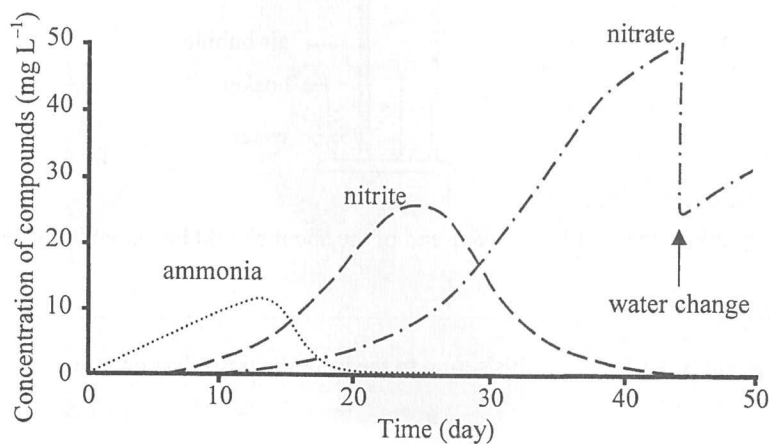
(2 marks)

- (ii) Is the foetus a boy or a girl? Explain your answer with reference to the photomicrograph.

(3 marks)

Answers written in the margins will not be marked.

5. Knowing that the nitrogen cycle is important to the healthy growth of fish, Tom purchased testing kits to monitor the levels of ammonia, nitrite and nitrate in his aquarium. He kept the aquarium aerated with an air pump. The graph below shows the changes in the concentrations of the three compounds over 50 days:



- (a) Name the process that led to the changes shown in the graph. (1 mark)

- (b) After three weeks, Tom observed a high concentration of nitrite in the water and the fish were showing symptoms of nitrite poisoning. Therefore, Tom added a suspension of a bacterial culture to the water to lower the nitrite concentration. Suggest the type of bacteria he added and explain how it lowered the nitrite concentration. (2 marks)

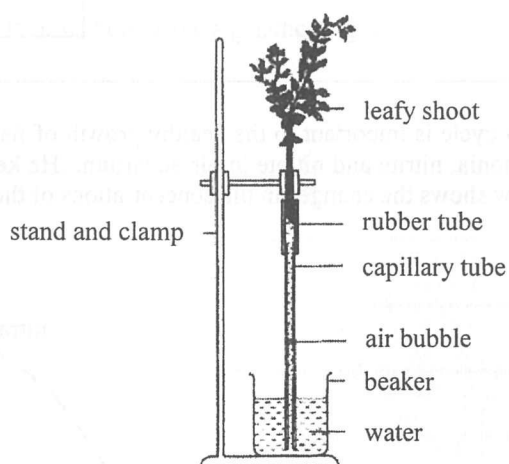
- (c) To prevent the accumulation of nitrate, Tom regularly replaced some water in the aquarium with fresh water. What else could Tom do to lower the nitrate concentration? (2 marks)

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6. The following set-up can be used to determine the transpiration rate of a leafy shoot:



- (a) In setting up this experiment, the lower end of the shoot should be cut under water. Why? (1 mark)

- (b) Give *one* assumption for using this set-up to measure the transpiration rate. (1 mark)

- (c) Explain how the transpiration rate will change if the fan placed near the shoot is switched on. (4 marks)

- (d) The photomicrograph below shows the appearance of the surface of a leaf during daytime:



- (i) In terms of sub-cellular structure, state **two** differences between cell A and cell B. (2 marks)

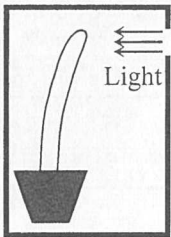
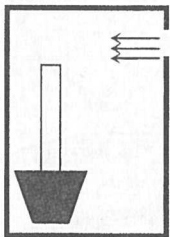
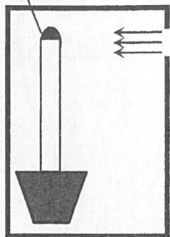
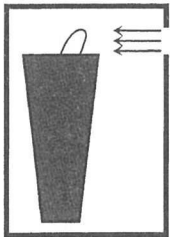
- (ii) What will happen to the size of C at night? Discuss the functional significance of this change. (3 marks)

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7. In 1880, Darwin conducted an experiment to investigate the phototropism of plants. He placed some coleoptiles in dark boxes, each with a hole at one side to allow light to pass through. The results after various treatments of coleoptiles are shown in the diagrams below:

Set-up	I	II	III	IV
Treatment of the coleoptile	Intact coleoptile	Tip removed	Opaque cap placed on the tip	Buried in soil with tip exposed
Result	 Growth with bending	 No growth and no bending	 Growth without bending	 Growth with bending

- (a) From the results of the experiment, which part of the coleoptile is responsible for detecting unilateral light? Support your answer with reasons. (3 marks)

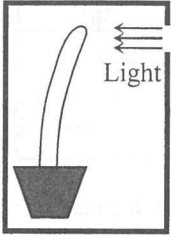
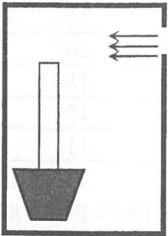
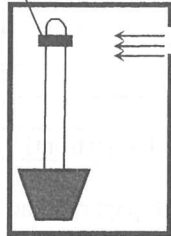
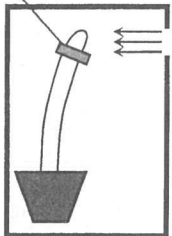
- (b) Explain why it is necessary to have set-up III in the experiment. (1 mark)

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- (c) In 1913, Boysen-Jensen performed some other experiments to study the nature of the signal transmission involved in phototropism. The diagram below shows his experimental set-ups:

Set-up	A	B	C	D
Treatment of the coleoptile	Intact coleoptile	Tip removed	Cut tip placed on a mica block	Cut tip placed on an agar block
Result	 Growth with bending	 No growth and no bending	 No growth and no bending	 Growth with bending

What conclusion can be drawn from Boysen-Jensen's experiment?

(3 marks)

- (d) Which of the following statements about the nature of science are demonstrated in the above historical events? Put a '✓' in the space next to the statement and provide an explanation. The first one is an example for your reference. (4 marks)

Statement		Evidence from the historical events
Doing science requires creativity and imagination.	✓	Both Darwin and Boysen-Jensen used innovation and imagination to design their experiments.
Science is socially and culturally embedded.		
Science is based on evidence.		
Science knowledge is tentative and dynamic.		

Answers written in the margins will not be marked.

8. A study was carried out to investigate the distribution and abundance of animal species A and B on a rocky shore. The table below shows the results obtained from the back of the shore to the waterfront:

Distance from the back of the shore (m)	Abundance (number of individuals m ⁻²)	
	Species A	Species B
1	10	0
2	25	2
3	40	8
4	38	10
5	20	20
6	18	35
7	15	33
8 (waterfront)	11	28

- (a) Using the graph paper on the opposite page, plot a graph to show the results. (5 marks)
- (b) Based on the information in the graph, suggest which species would have a higher tolerance to desiccation. Explain your answer. (4 marks)

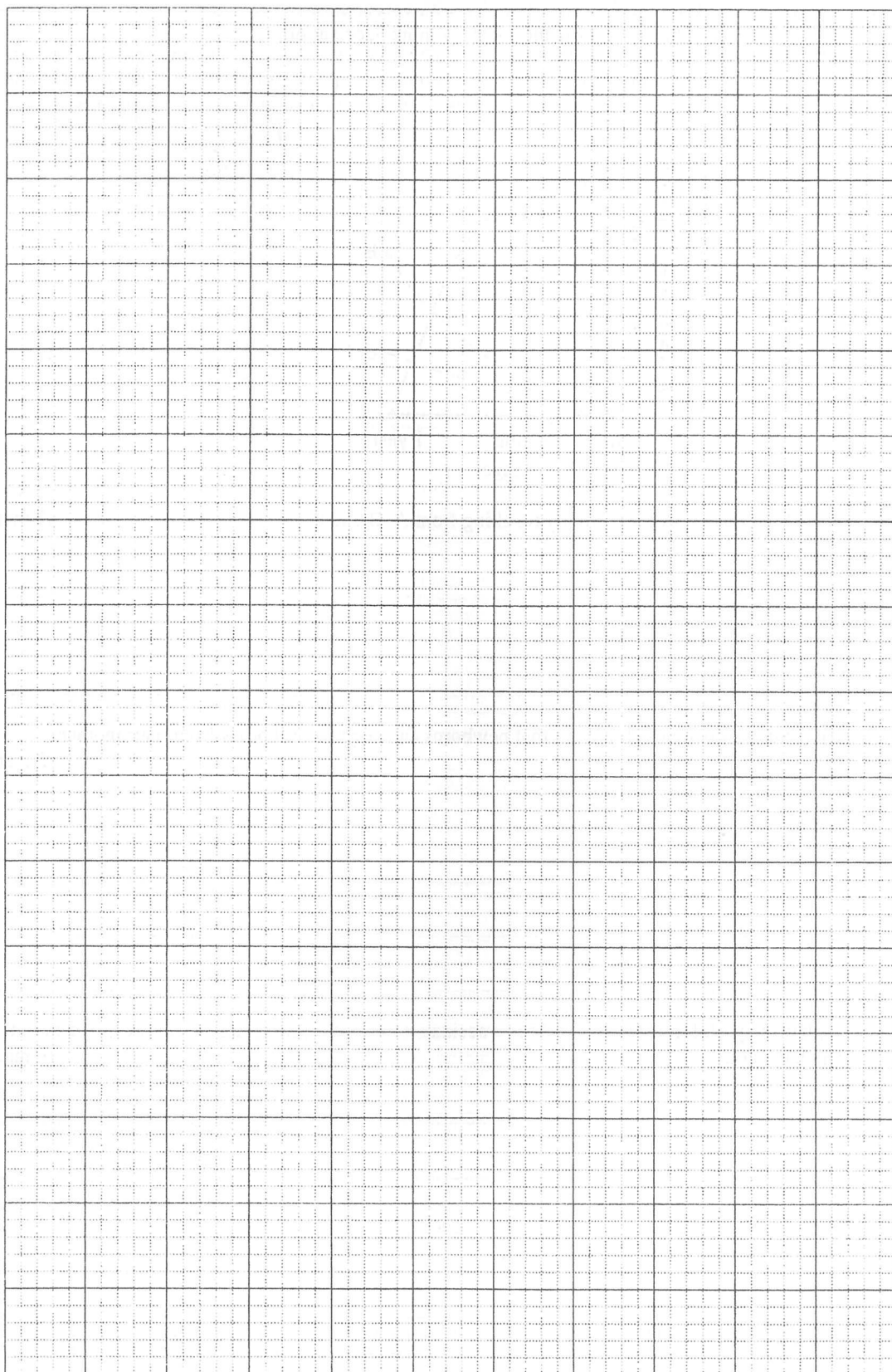
- (c) Briefly describe the sampling procedure used to obtain the above data. (3 marks)

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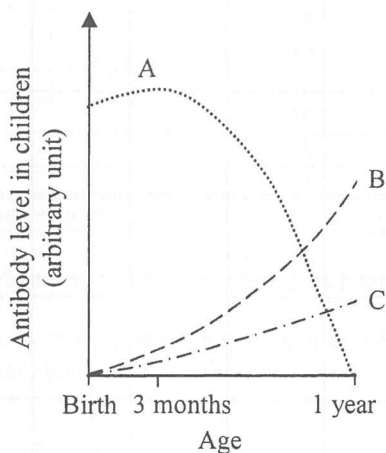
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9. (a) The following graph shows the change in levels of antibodies in children's bodies:
- (A) antibodies from mother
 - (B) children's own antibodies with vaccination
 - (C) children's own antibodies without vaccination



- (i) State the types of immunity resulting from A, B and C. (3 marks)

A: _____

B: _____

C: _____

- (ii) Suggest *two* possible ways that newborns can acquire antibodies from their mother. (2 marks)

- (b) Explain why children who have been vaccinated against diseases are better protected than those who have not. (4 marks)

For the following question, candidates are required to present their answer in essay form. Criteria for marking will include relevant content, logical presentation and clarity of expression.

- *10. Proteins serve different functions in our body and their functional role is closely related to their conformation (shape). Describe how protein molecules can have different conformations and explain how the different conformations enable them to carry out different functions. (11 marks)

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* In the live paper, the first word of the second line was misprinted as ‘conformaton’. As the meaning of the word – shape – has been given in brackets and the same word was correctly printed twice in the subsequent sentence, this misprint did not affect candidates’ understanding of the question. The question in this Examination Report and Question Papers has been retified.

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END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

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